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COURSE NAME : CLOUD COMPUTING FOR BIG DATA ANALYSIS

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Assignment - 1

Evolution of cloud Computing ?

Cloud Computing has evolved from traditional Computing to modern - on - demand cloud Services

Stages of Evolution:

1. Mainframe Computing (1960s - 1980s)

* One Central Computer Serves many users
* Expensive and limited access
* Used mainly by Large Organizations

2. Client - Server Computing (1980s - 1990s)

* clients requested Services from Servers
* Improved resource Sharing
* Easier application management

3. Grid Computing:

* Multiple Computers worked together

Increased processing power
Used for Scientific research.

4. Virtualization

* Multiple Virtual machines run on one physical server
* Better hardware utilization
* Reduced infrastructure cost

5. Cloud Computing:

* Services delivered over the Internet
* Users pay only for what they use
* Supports Iaas, Paas, and Saas

Impact on Traditional IT

* Lower infrastructure cost
* faster deployment
* high Scability

* Better disaster recovery

* Remote access from anywhere

* Improved flexibility and Collaboration

Conclusion:

Cloud Computing transformed traditional IT into a flexible, Scalable, and Cost-effective environment that supports modern businesses

2. Hardware Evolution in cloud Computing:

Answer:

Modern cloud Computing depends on Powerful hardware to deliver high Performance.

Hardware Innovations:

1. GPUs (Graphics Processing Units)

* Accelerate AI and machine Learning

* Process many tasks Simultaneously

* Improve deep learning Performance.

3. INTERNET SOFTWARE Evolution:

Answer:

Internet Software has evolved to support modern cloud Applications

Major Developments:

1. Web Servers

* Appache
* Nginx
* TIS

They host websites and cloud applications

2. APIs (Application Programming Interfaces)

* Enable Communication between applications
* Simplify Service integration
* Support cloud-based Services

* Faster data transfer
* Low Latency
* Better cloud Communication.

Benefits:

* High Performance
* faster Processing
* Better Scalability
* Reduced Power Consumption
* Improved reliability.

Conclusion:

Advanced hardware enables cloud platforms to handle AI, big data, Simulations, and other demanding Workloads efficiently.

3. HTTP and HTTPs

* HTTP transfers web data
* HTTPS Provides Secure Communication using encryption

4. RESTful Web Services:

* Lightweight
* Easy to develop
* Widely used in cloud applications

5. Web Sockets

* Enable real time Communication
* Used in chat application and online gaming

4: Server Virtualization:

Server Virtualization allows multiple Virtual machines to run on one physical Server

How it works

A hypervisor divides one Physical Server into multiple Servers

Benefits:

- Better Resource utilization
- Cost Reduction
- Scalability
- Easy back-up & Recovery
- Isolation.

Cloud Examples:

Vmware ESXi, Microsoft hyper V, Kvm

Oracle Virtual Box

Conclusion:

Server Virtualization improves resource utilization, reduces Cost, and Provides the Scalability

5. Web Services Overview

Web Services allow different applications to Communicate over the Internet

Importance in cloud Computing:

1. API Integration:

* Connects different applications
* Enables Payment gateways, maps and authentication

* 2. Service Interoperability:

* Applications build on different platforms work together
* Supports data sharing.

3. Automation:

* Reduces manual work

* Improves efficiency

4. Scalability:

ApI's handle increasing number of users

Cloud Services expand automatically

5. Security

* Authentication using OAuth and ApI Keys

* Data Encryption using HTTPS

REAL WORLD EXAMPLE:

* Payment ApI

* SMS ApI

* EMAIL ApI

* BANK ApI